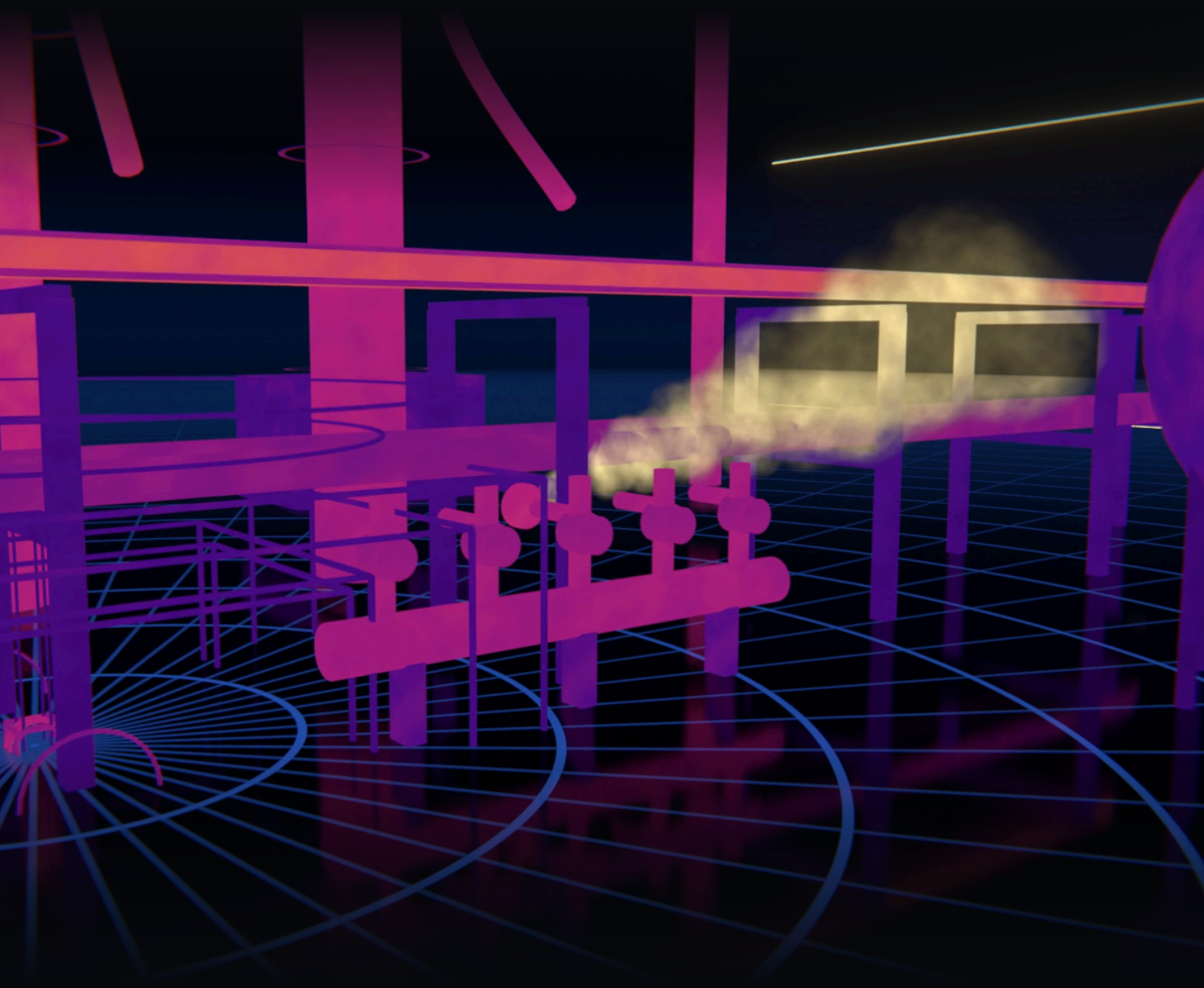
GLD NEXT · DEVELOPMENT CONCEPT

Optical Gas Imaging

Seeing invisible hydrocarbon leaks, from point to area.



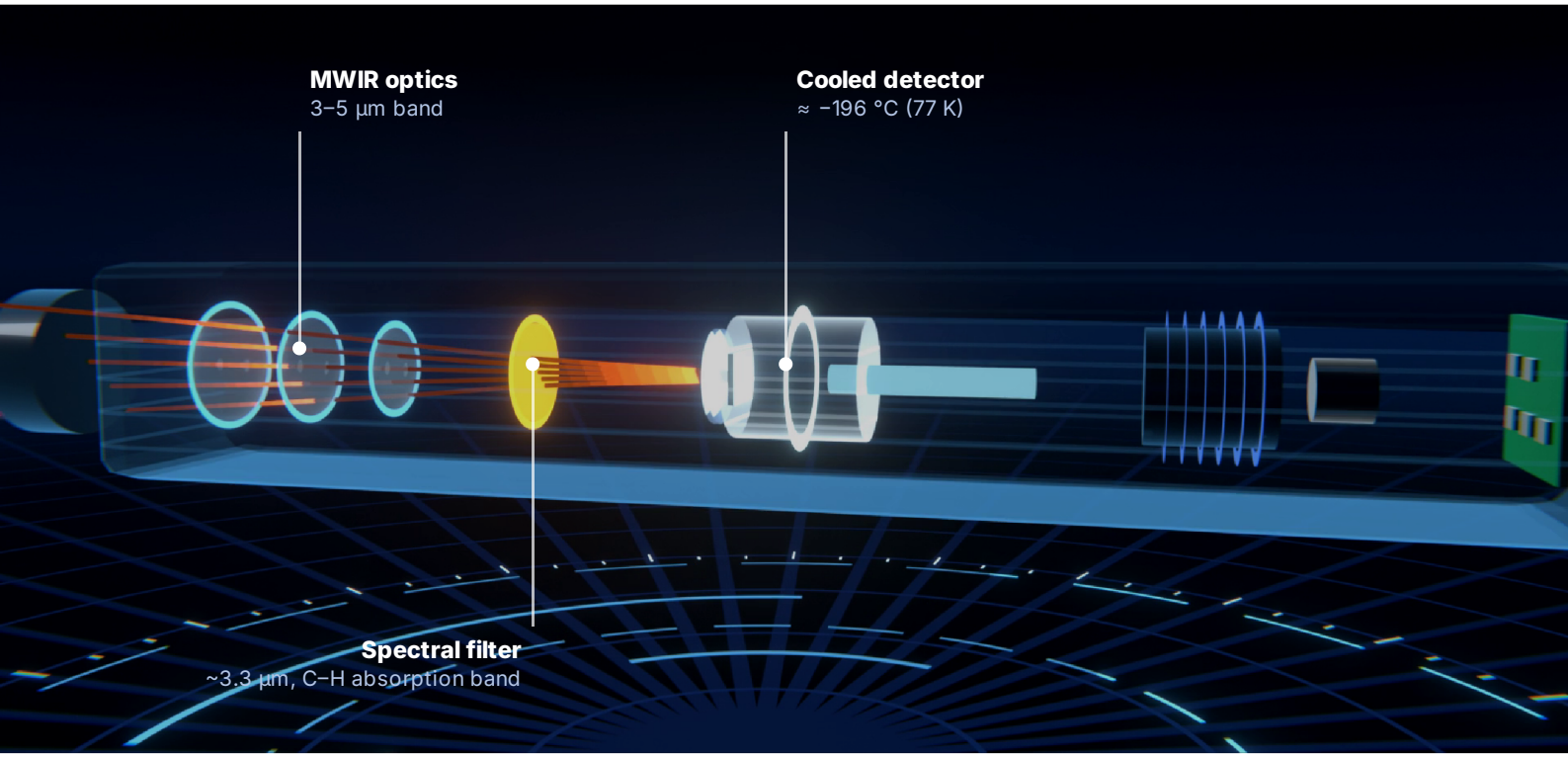
PT Pertamina Patra Niaga (Reliability Function)
LAPI Ganesha Utama · Institut Teknologi Bandung

3D illustration

THE PRINCIPLE

Invisible gas. Made visible.

Carbon-hydrogen bonds absorb infrared radiation near 3.3 µm. A camera with a narrow spectral filter on that band renders the gas as contrast against the background.



3D illustration of a generic cooled MWIR camera.

MWIR band

Mid-Wave Infrared, 3–5 µm, where hydrocarbons absorb radiation.

Narrow filter

Passes only the C–H absorption wavelength.

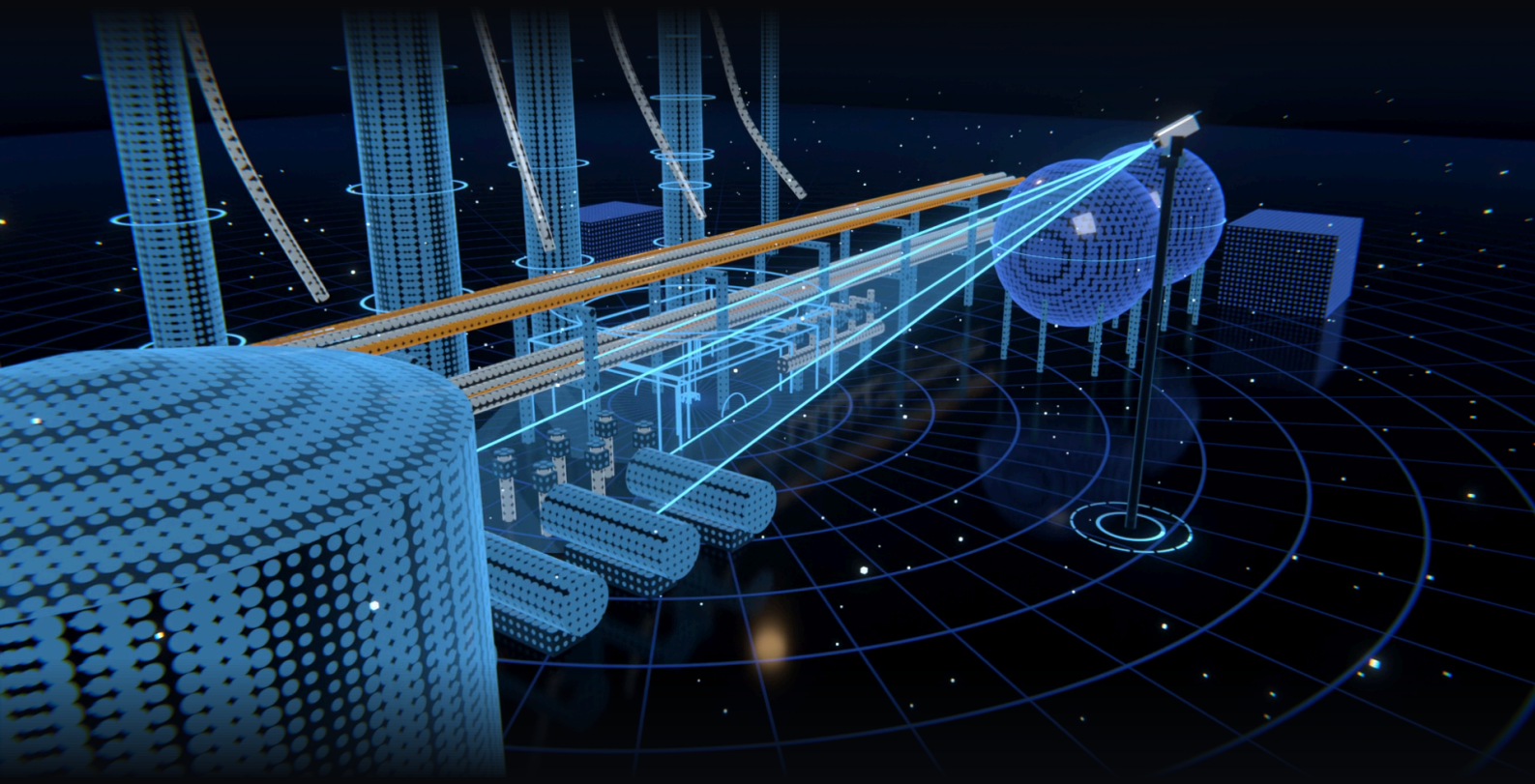
Cryogenic detector

Cooling reduces detector noise, improving sensitivity to weak infrared gas signatures.

THE SYSTEM

From point to area.

A pan-tilt camera on a ~10 m pole patrols critical sectors: pipe runs, manifolds and valve clusters. Target range ~100 m.



MWIR camera → Edge server · AI plume segmentation → Dashboard GLD + OGI

GLD + OGI, one system

GLD	Which gas & how concentrated
OGI	Where it is & which way it spreads
Both	**Highest response priority**

Confirmed alarms

An alarm is raised only when the plume is detected consistently over consecutive frames, suppressing flares and heat reflections.

3D illustration of the system concept.

COMPLEMENTING GLD

Complementing, not replacing.

3–5 µm
Mid-Wave Infrared band

−196 °C
Detector operating temperature

±100 m
Target visualisation range

TARGET SPECIFICATIONS

Detector	Cooled MCT, 640 × 512 pixels
Frame rate	100 Hz
Mounting	~10 m pole, pan-tilt, preset sector patrol
Intelligence	Plume segmentation on edge server, multi-frame confirmation
Validation	Controlled gas release tests
Limitations	The proposed OGI system is optimized for hydrocarbon detection near 3.3 µm and is not designed to detect H_2, CO, or H_2S; can be addressed by the GLD system.

ROADMAP

Modular
Designed for replication across Pertamina refineries.

Fugitive emissions
Supports fugitive emission documentation.

Predictive maintenance
Radiometric thermography: heater hot spots, damaged insulation, steam traps.

Next-stage development concept, not part of the current GLD scope. Specifications are design targets; all images are 3D illustrations.

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